

Competitive Dynamic Pricing · Nash Equilibria

WEEK 3 OF 8

Multi-Agent RL · Oligopoly Simulation · Collusion Detection

PHASE 2 — RULE-BASED AGENTS

Week 3: Rule-Based Baseline Agents — Build Tit-for-Tat, Always-Nash, Always-Collude, and Random agents before any RL code. These baselines validate the environment AND give the RL agent something meaningful to beat. Without baselines, there is no way to judge whether the RL agent learned anything useful.

Priority key:

MUST WATCH / READ

RECOMMENDED

OPTIONAL

■ = estimated time | pp. = page range

■ YouTube & Interactive Videos

- **Game Theory — Cooperation and the Prisoner's Dilemma**
Crash Course Economics
youtube.com/watch?v=qcLZMYPdPH4
Crash Course's clearest episode on repeated games and cooperation. Great group-watch opener before coding Tit-for-Tat. Fun, fast, and accurate.

MUST WATCH
■ 10 min
- **Prisoner's Dilemma and Tit-for-Tat Strategy**
William Spaniel — Game Theory 101
youtube.com/watch?v=uvmz5E75ZIA
Best YouTube series for applied game theory. Explains exactly why Tit-for-Tat wins Axelrod's tournament. Watch before coding the agent.

MUST WATCH
■ 8 min
- **Repeated Games and the Folk Theorem**
William Spaniel — Game Theory 101
youtube.com/watch?v=D4pnin-QGwA
Why firms can sustain collusion in repeated interactions — the theoretical basis for why Always-Collude is tempting but unstable. Directly relevant to Week 6.

MUST WATCH
■ 10 min
- **The Evolution of Trust (Interactive Game)**
Nicky Case
ncase.me/trust
Play ALL three acts. The single best intuition-builder for cooperation and defection. Students who play this write far better strategy classes.

MUST WATCH
■ 30 min
- **Multi-Agent Systems — Introduction**
Didactic AI
youtube.com/watch?v=4hx_8H4bnB8
Introduces multi-agent simulation, round-robin tournaments, and emergent strategy — the exact framing for the tournament students build this week.

RECOMMENDED
■ 22 min
- **Python Agent-Based Modelling with Mesa**
sentedex
youtube.com/watch?v=xcsVtBvendY
Structuring agents as Python classes with the Mesa library. Reference the class structure for rule-based agent implementations.

RECOMMENDED
■ 25 min
- **What Game Theory Reveals About Life, the Universe, and Everything**
Veritasium
youtube.com/watch?v=mScpHTli-kM
Conceptual and engaging — good for motivation and group discussion. Not technically deep but very accessible and well-produced.

OPTIONAL
■ 20 min

■ Papers, Books & Blogs

- **The Evolution of Cooperation — Ch. 1 & 2**
Robert Axelrod (1984)
[Library or PDF search: 'Axelrod Evolution of Cooperation'](#)
Origin of Tit-for-Tat. Ch. 1–2 explain Axelrod's tournament and why TFT dominates — essential before coding the agent.

MUST READ
■ ~50 min
pp. 3–54

Effective Choice in the Prisoner's Dilemma Axelrod — Journal of Conflict Resolution, 1980 doi.org/10.1177/002200278002400101 The original tournament paper. Note the 4 properties (nice, retaliating, forgiving, clear) that make a strategy robust in repeated play.	MUST READ ■ ~25 min pp. 3–25
→ Simulating Competitive Markets with Rule-Based Agents in Python Towards Data Science towardsdatascience.com → search title Direct implementation guide. Covers strategy classes, tournament loops, and result visualisation — saves significant setup time.	RECOMMENDED ■ ~20 min Web article
AI, Algorithmic Pricing and Collusion — Section 2 (re-read) Calvano et al. — QJE 2020 doi.org/10.1093/qje/qjz036 Re-read with rule-based agents in mind. Note how the authors frame baselines before introducing RL agents — mirror this in your final report structure.	OPTIONAL ■ ~20 min pp. 6–12

■ Common Pitfalls to Watch — Week 3

■ Always-Collude appears to 'win' short-term — run 1,000+ rounds for defection incentives to emerge properly.

■ Not seeding the Random agent — results will be irreproducible. Always set and log all random seeds.

■ Skipping tournament logs — raw results tables are needed for the final report's baseline comparison section.

■ Videos (must)	■ Videos (rec.)	■ Reading (must)	■ Total (must only)	■ Total (all items)
58 min	47 min	75 min	~2.5 hr	~5.5 hr